

Computer Networks I

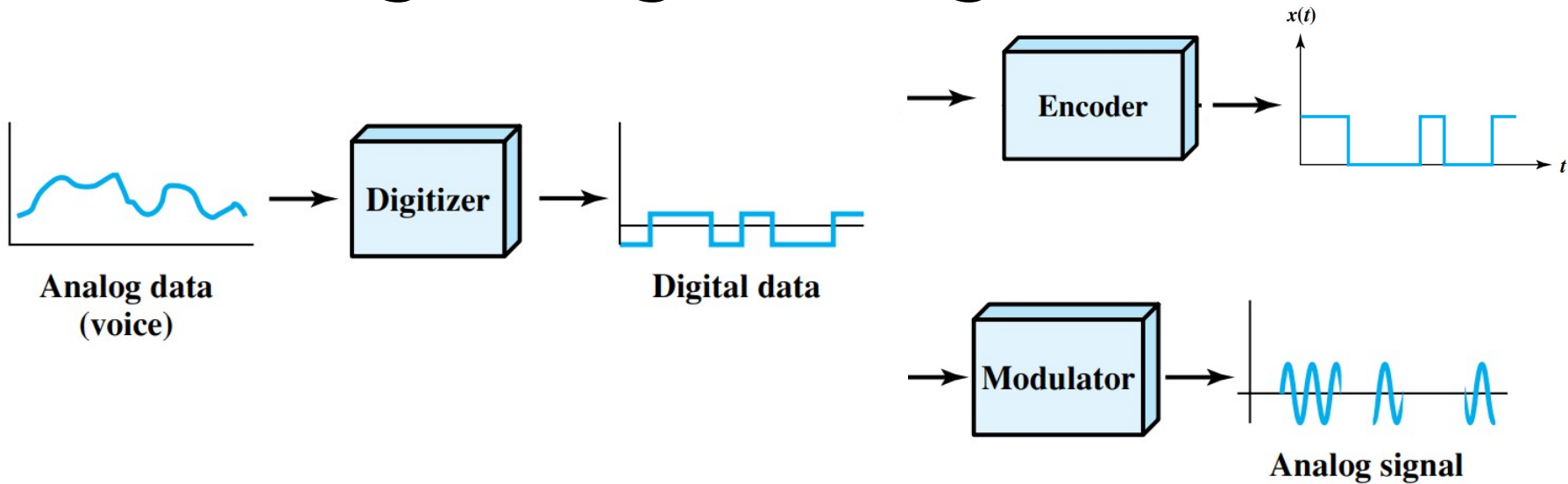
Signal Encoding Techniques (Analog to Digital)

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Digitizing Analog Data

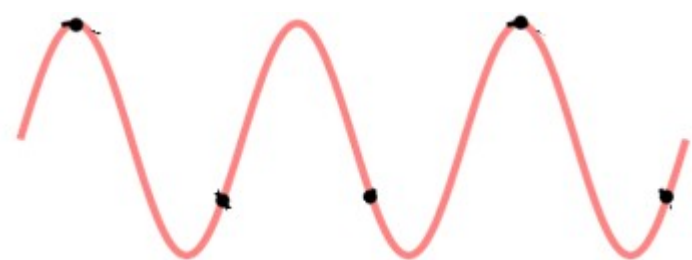


- Once analog data is converted to digital data:
 - The digital data can be encoded into digital signals
 - The digital data can be modulated into an analog signal
- Two principle techniques:
 - Pulse Code Modulation (PCM)
 - Delta Modulation (DM)

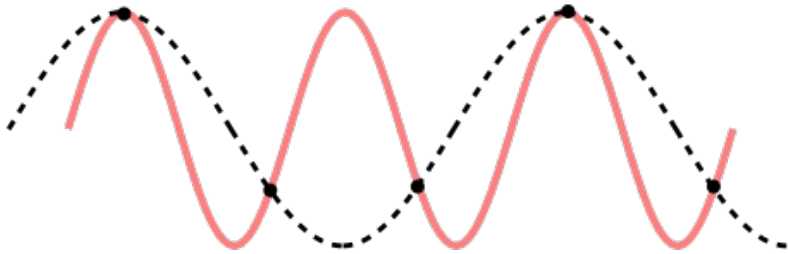
Nyquist Sampling Theorem



Src: <https://commons.wikimedia.org/wiki/File:CPT-sound-nyquist-theorem-raw.svg>



Src: <https://commons.wikimedia.org/wiki/File:CPT-sound-nyquist-theorem-1.5percycle.svg>



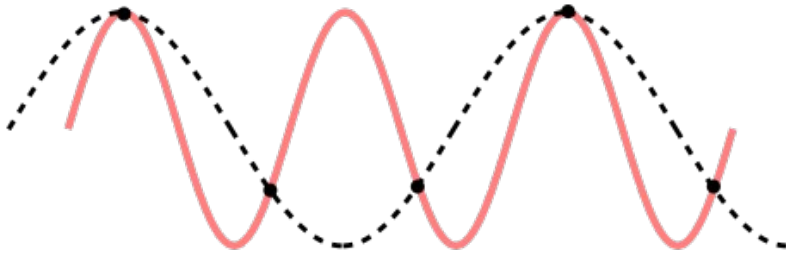
Src: <https://commons.wikimedia.org/wiki/File:CPT-sound-nyquist-theorem-1.5percycle.svg>



Src: <https://commons.wikimedia.org/wiki/File:CPT-sound-nyquist-theorem-2percycle.svg>

Nyquist Sampling Theorem

- **Sampling theorem:** If a continuous time signal is sampled, then it can be reconstructed from the samples, if the sampling rate must be **at least 2 times the highest frequency** contained in the signal

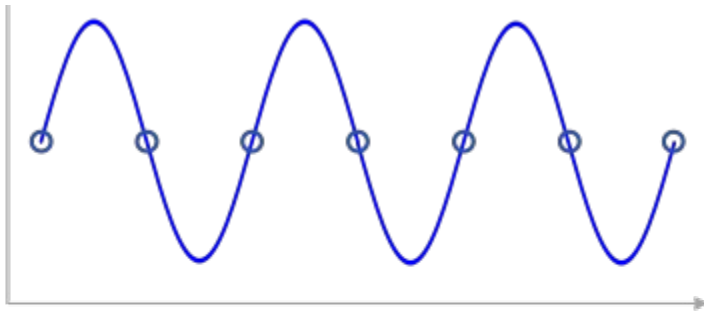


Src: <https://commons.wikimedia.org/wiki/File:CPT-sound-nyquist-theorem-1.5percycle.svg>



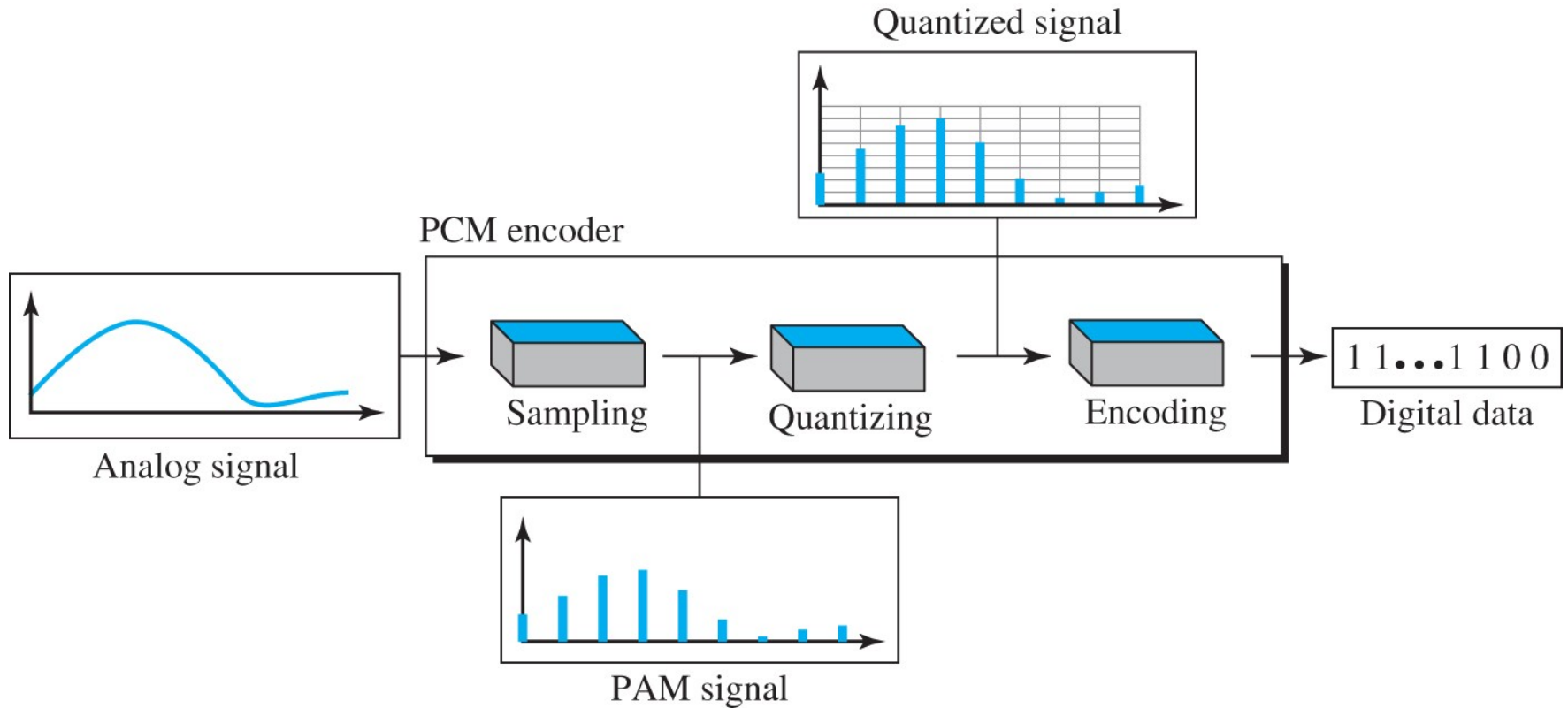
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Is it sufficient to keep $f_s = 2f_{\max}$?

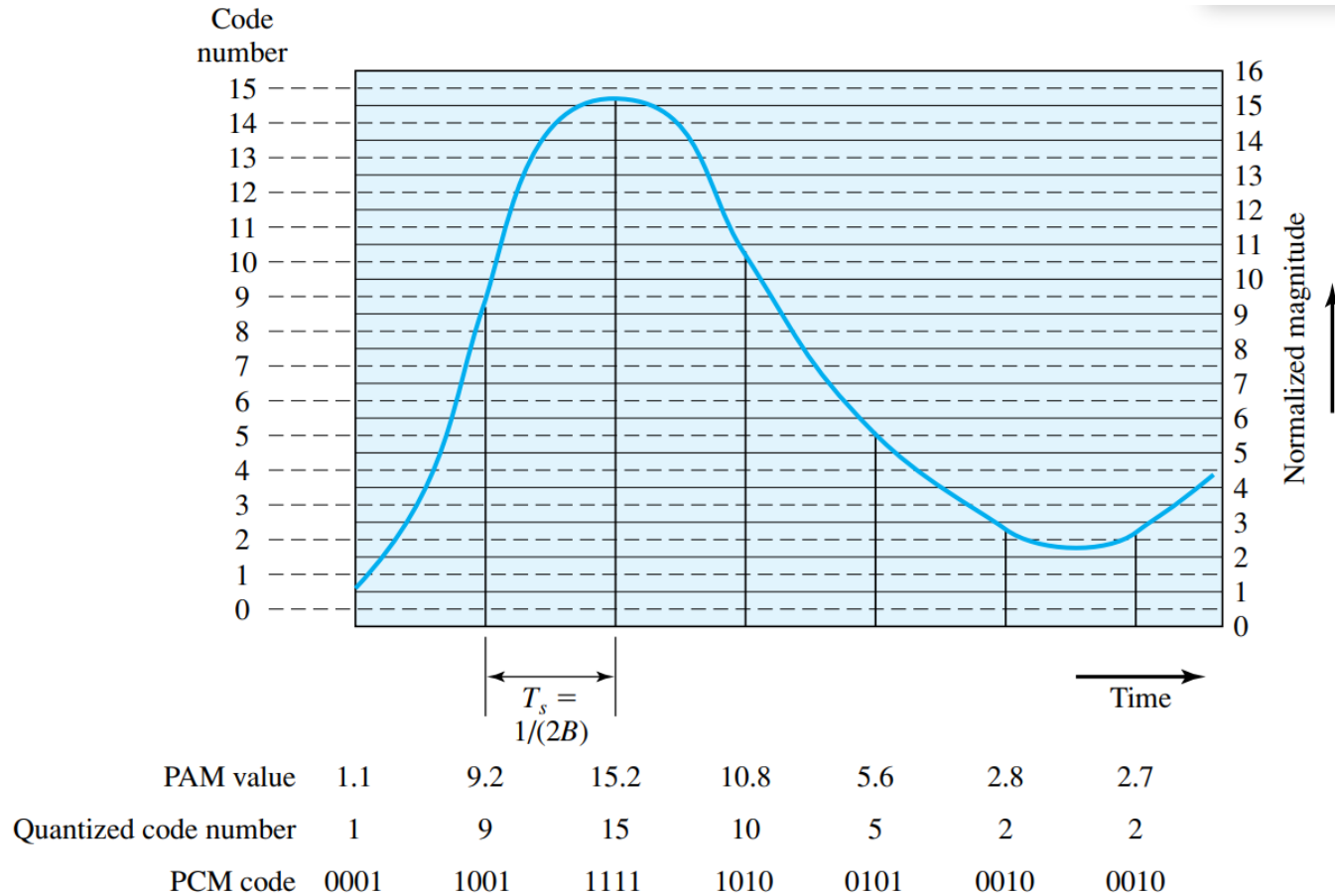


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https://commons.wikimedia.org/wiki/File:Sampling_theorem_li mit.svg

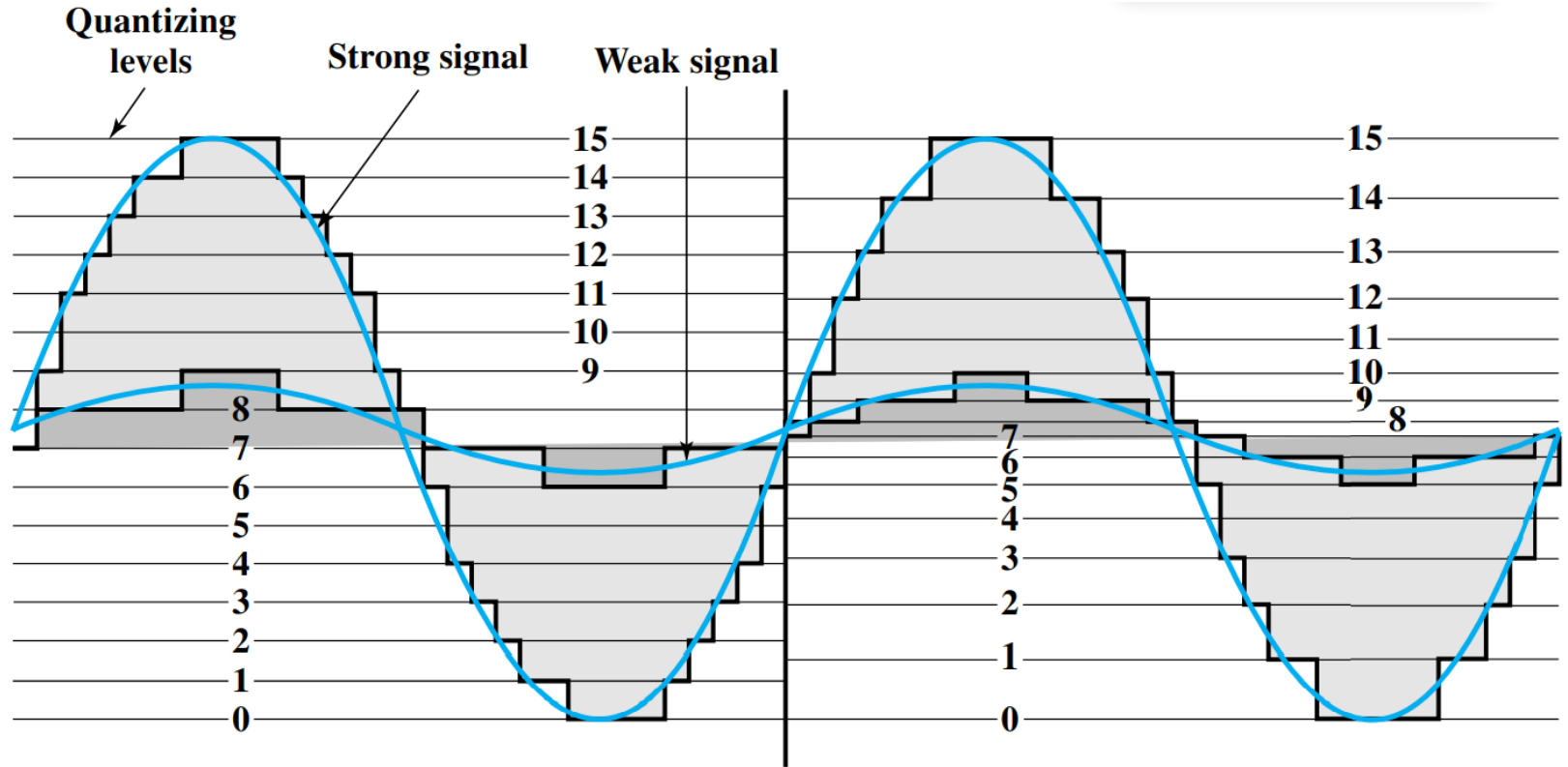
Pulse Code Modulation Block Diagram



Pulse Code Modulation



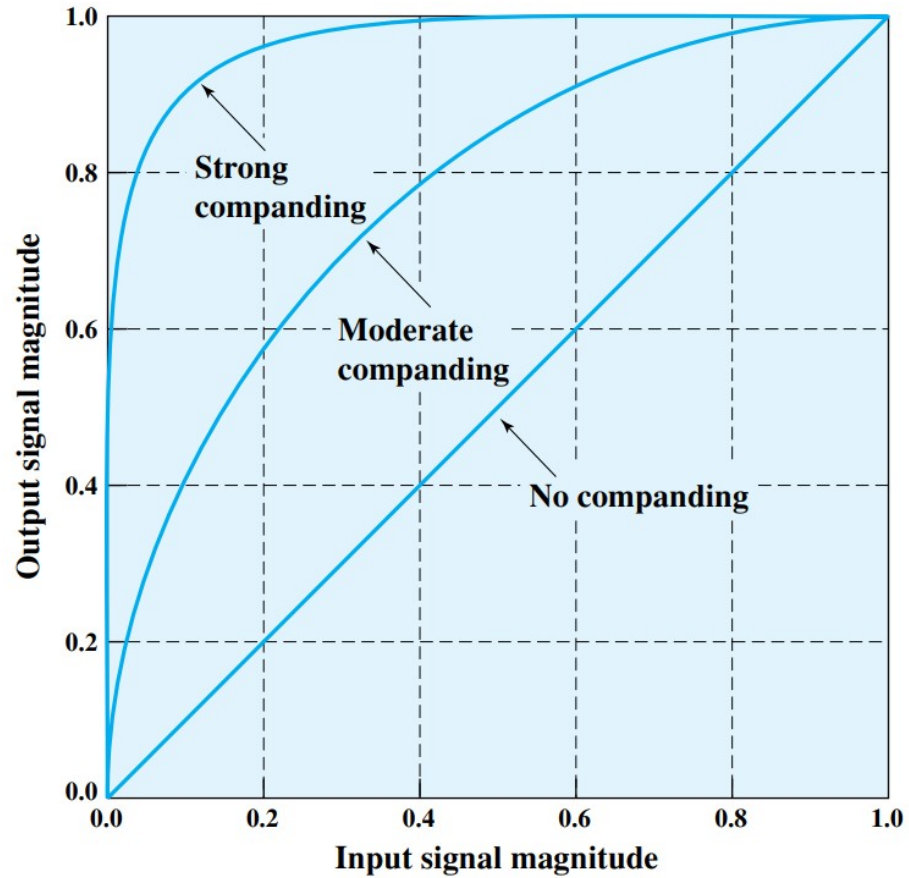
Nonlinear Coding



(a) Without nonlinear encoding

(b) With nonlinear encoding

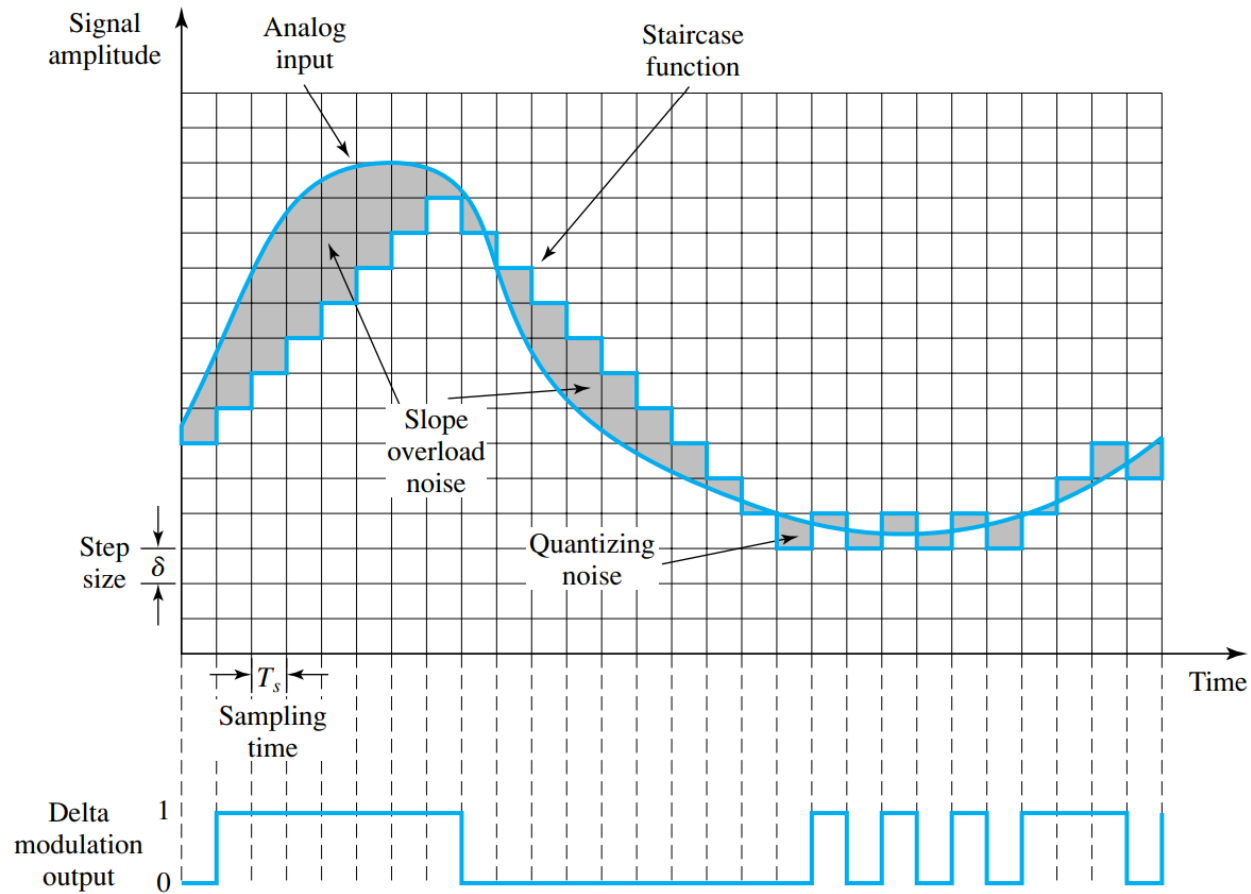
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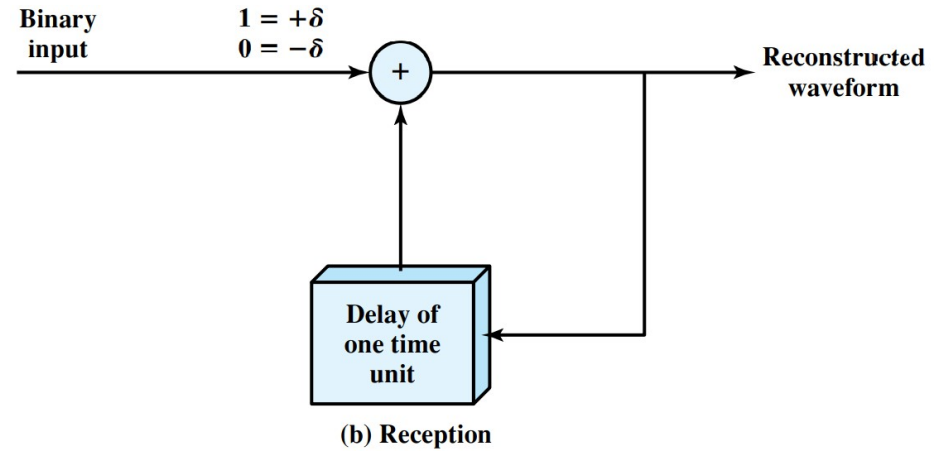
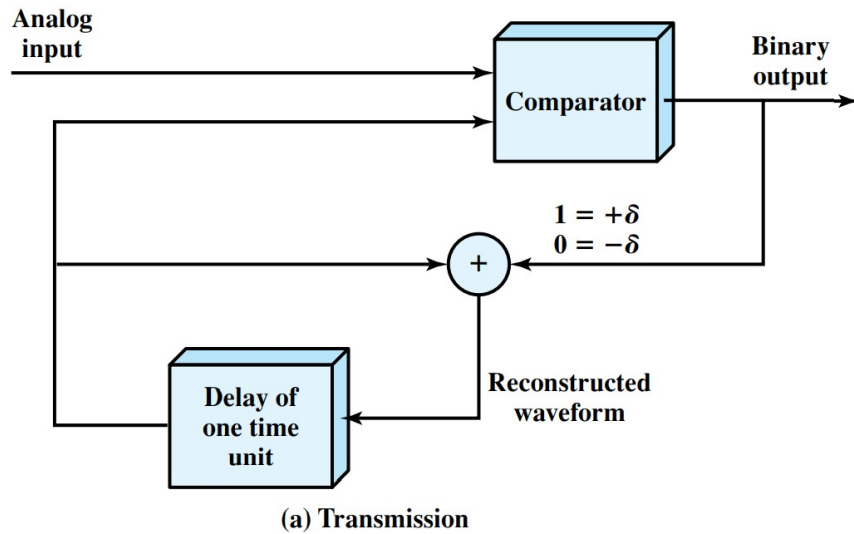
Pulse Code Modulation

- A voice signal occupies a bandwidth of 4 KHz (assume that this is the maximum frequency). Now good voice reproduction via PCM can be achieved with 128 quantization levels. Then what will be the minimum data rate for the PCM encoded digital data?
 - 128 quantization levels $\rightarrow$ 7 bit coding
 - The maximum frequency is 4khz $\rightarrow$ the sampling rate is atleast 8000 samples/sec
 - Therefore, the datarate will be atleast $8000 \times 7 = 56\text{kbps}$
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Delta Modulation



Delta Modulation Operation



PCM versus Delta Modulation

- DM has simplicity compared to PCM
 - DM requires lower transmission bandwidth
 - DM has poorer SNR as compared to PCM
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Summary

□ Signal encoding techniques (Analog data → Digital signals):

- Nyquist Sampling Theorem
 - Pulse Code Modulation
 - Delta Modulation
 - Comparison of Pulse Code Modulation and Delta Modulation
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